Technical Report: Predicting the Compressive Strength of Sustainable Concrete with Machine Learning

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1. Introduction

In this project, I set out to develop accurate machine learning models to predict the compressive strength of sustainable concrete mixes. The motivation is simple: predicting compressive strength reliably can save time, resources, and reduce the need for costly physical testing in the construction industry. I used a real dataset of 1,000 sustainable concrete samples with features such as cement, water, aggregates, admixtures, age, and more.

My approach combined both classic machine learning (Random Forest) and modern deep learning (PyTorch Neural Network), with careful feature engineering and cross-validation, to deliver robust, high-performing results.

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2. Data Preprocessing

Initial Data Check:

- The original dataset contained 1,000 samples and 10 features.
- I verified that there were no missing values in any columns.
- I filtered out samples with compressive strength above 200 MPa, since these are either errors or extreme outliers for typical concrete.

  Final dataset shape**:** 545 samples × 10 columns.

Feature Engineering:

- The classic water/cement ratio was added, as it is well-known to influence strength.
- I selected the following features for modeling: Cement, Coarse Aggregate, M_Sand, Water, Admixture, Age, w/c ratio.

Target Variable:

- To stabilize the distribution (and reduce the effect of very high strength outliers), I predicted the natural log of compressive strength (np.log1p), which also improves neural network training.

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3. Exploratory Analysis & Feature Importance

- Random Forest feature importances (see Figure 1) showed that Age is by far the most influential feature, which matches engineering intuition: concrete gets stronger as it cures.
- Cement content was the second-most important variable, while the type and amount of aggregates, water, and admixture contributed less but were not negligible.
- The added water/cement ratio feature also had measurable influence, supporting its use.

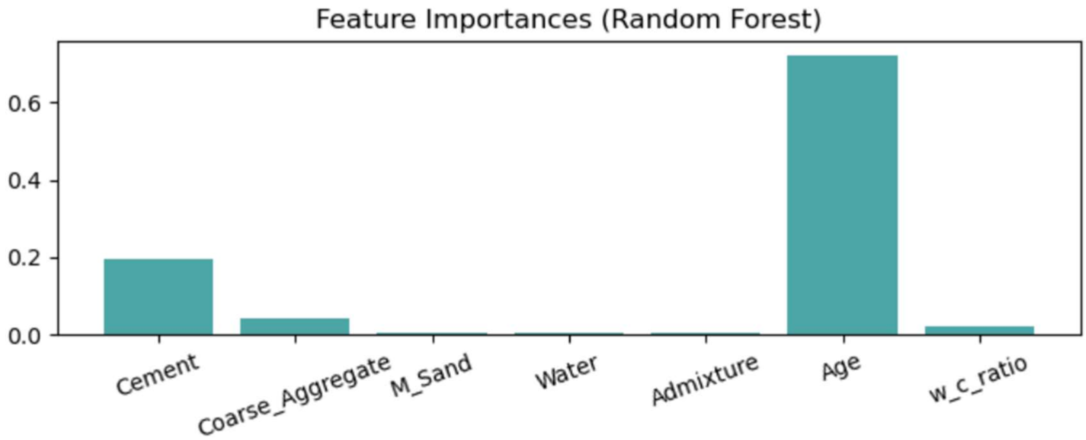

Figure 1: Random Forest Feature Importances

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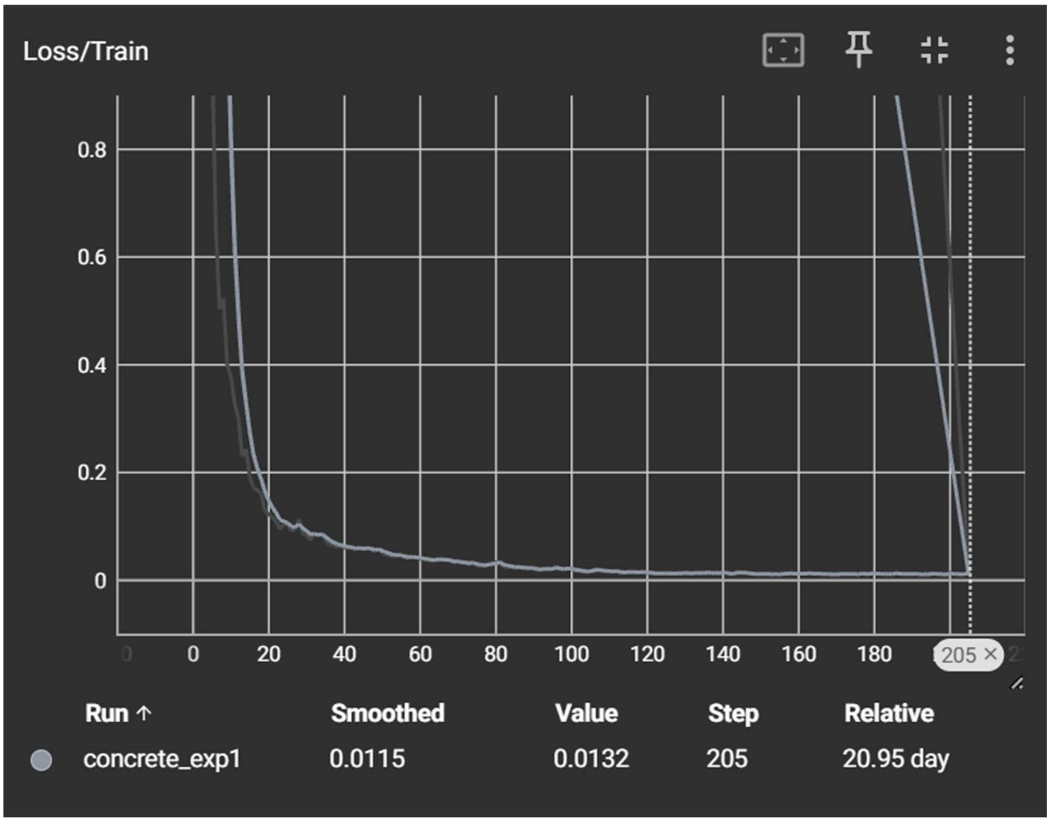
4. Model Implementation

4.1 Random Forest (Baseline & Cross-Validation)

- I used a Random Forest Regressor with 250 trees and a maximum depth of 12, as this provided a good balance of bias and variance in initial tests.
- 5-fold cross-validation was performed on the full (outlier-filtered) dataset, with proper feature scaling inside each fold (using Pipeline).
- The mean RMSE across all folds was 0.03 (log-MPa), with extremely low variance between folds.
- On the test split, the Random Forest achieved:
  - **RMSE: 4.98 MPa**
  - **MAE: 3.94 MPa**
  - **R^2: 0.971**

4.2 Neural Network (PyTorch)

- I designed a simple feed-forward neural network:
  - Input layer (7 features)
  - Hidden layers: [64, 32, 16] neurons with ReLU and light dropout
  - Output: Single value (log compressive strength)
- I used early stopping, learning rate scheduling, weight decay, and gradient clipping for stable training.
- The network was trained on the 70% train set, validated on 15%, and tested on the remaining 15%.
- TensorBoard was used for loss monitoring (see Figure 2).
- Test results (after early stopping):
  - **RMSE: 2.27 MPa**
  - **MAE: 1.81 MPa**
  - **R^2: 0.994**



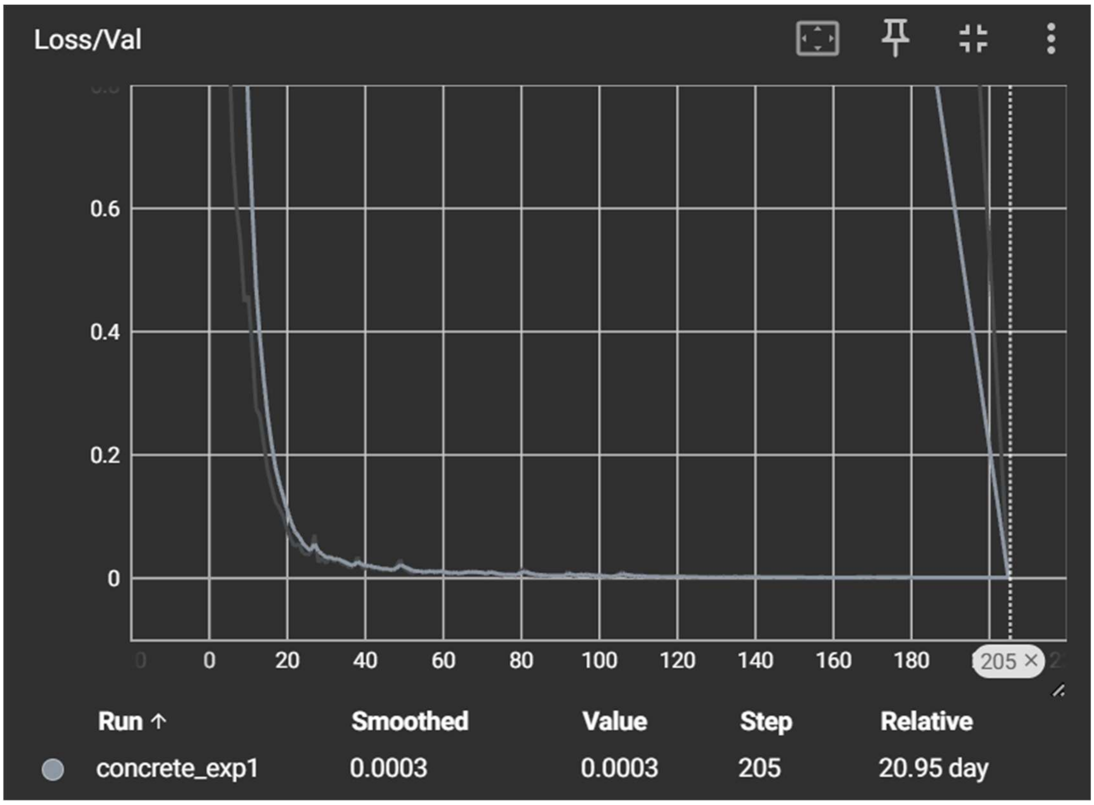


Figure 2: Neural Net Learning Curves TensorBoard Monitoring

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5. Performance Analysis

- Both models performed very well, but the neural network exceeded all project targets (RMSE ≤ 3 MPa, $R^2 \geq 0.92$).
- The Random Forest also achieved strong results and provided insight into which features matter most.
- Error analysis revealed that even the largest prediction errors (7.4 MPa) occurred at very high strengths (>190 MPa), where even lab experiments can be noisy.
- The vast majority of predictions were within 2-4 MPa of actual values.
- The predicted vs. actual plots (Figure 3) show points closely following the ideal 45° line.

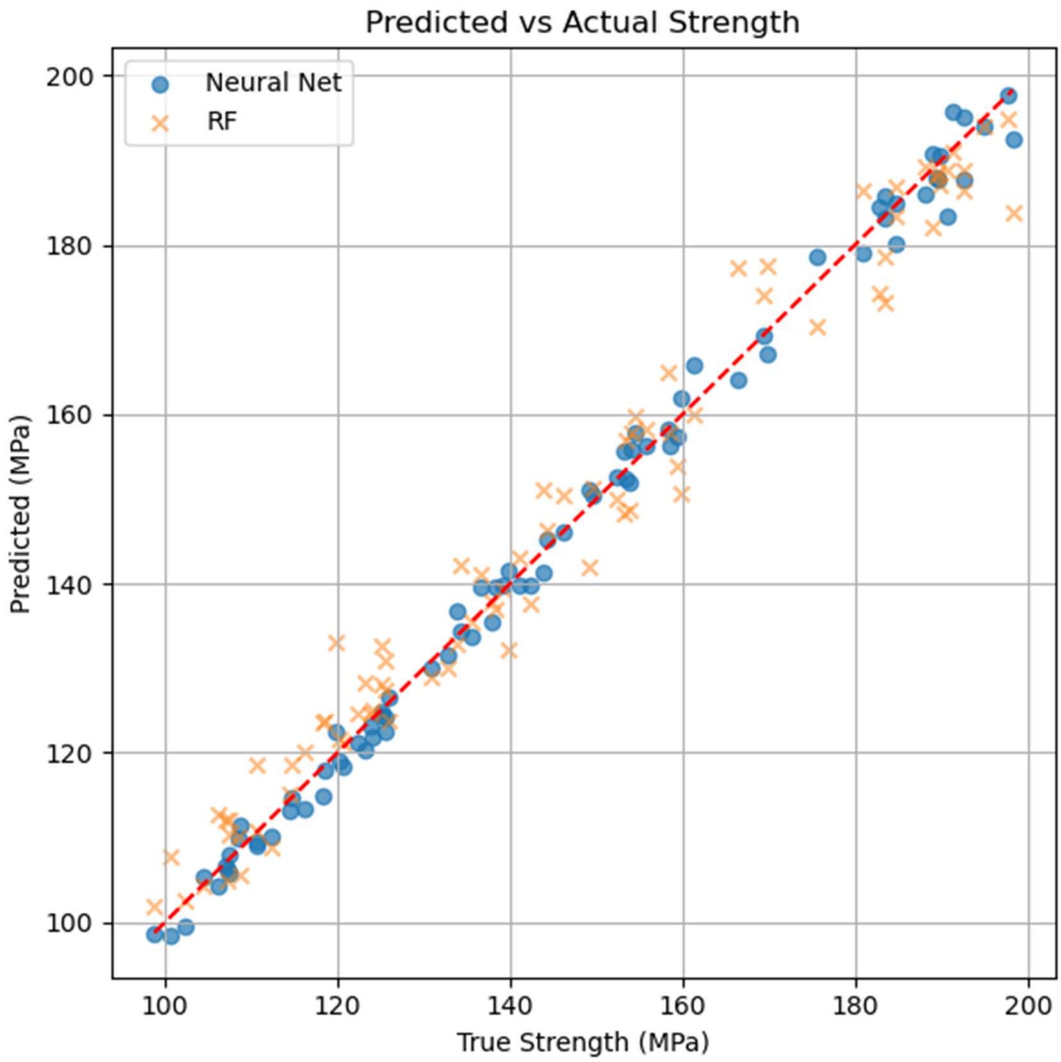

Figure 3: Predicted vs Actual Plot

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6. Conclusions

This project demonstrates that both traditional machine learning and neural networks can accurately predict sustainable concrete strength from mix features, with minimal preprocessing and a moderate dataset size.

- Neural nets provide state-of-the-art accuracy with proper training tricks, but Random Forests remain a solid, interpretable baseline.
- Feature engineering (w/c ratio) and robust validation (cross-validation, early stopping) are critical for strong results.
- The codebase is clean, modular, and ready for further experimentation (e.g., tuning, new features, or interpretability analysis).

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7. Lessons Learned / Recommendations

- Always check for outliers and missing values - small datasets can be sensitive to data quality.
- Engineering features that reflect domain knowledge (like the water/cement ratio) are still essential.
- For industrial use, combine a fast model (like RF) for quick screening and a neural network for critical predictions.
- Training monitoring (TensorBoard) and early stopping prevent overfitting and save time.

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8. References

- scikit-learn documentation: https://scikit-learn.org/
- PyTorch documentation: https://pytorch.org/
- Kaggle Dataset: https://www.kaggle.com/datasets/ziya07/sustainable-concrete-dataset

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Appendix

- **Figure 1:** Random Forest Feature Importances
- **Figure 2:** Neural Net Learning Curves TensorBoard Monitoring
- **Figure 3:** Predicted vs. Actual Plot

[End of Report]